

PRANAV PANDEY

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SUMMARY

Senior Applied AI Engineer with **7+ years** building production **Generative AI (GenAI) and ML systems** at scale, from **multi-agent LLM platforms**, **Retrieval-Augmented Generation (RAG)** architectures, and document pipelines (**100K to millions of documents**) to deep-learning models on **1M+ edge devices**. Strong **Python backend engineering** (FastAPI, REST APIs, distributed pipelines) and **system design** ownership across **agent orchestration**, **hybrid retrieval**, **knowledge graphs**, **LLM evaluation**, **fine-tuning**, and **multi-model inference**. Took AI products from zero to revenue (**\$500K in 3 months**), drove **70%+ efficiency gains**, cut costs (**\$84K/yr**); former startup **CTO/co-founder**.

EXPERIENCE

Senior Applied AI Engineer

July 2025 - Present

SpotDraft

India

Tech: Python, LangGraph, LangChain, LiteLLM, OpenAI, Claude (Anthropic), Apache Beam, Google Cloud Dataflow, Azure, GCP, FastAPI, Vector Databases, Knowledge Graphs

- Lead engineer for **Sidebar (AI agents for legal teams)**, SpotDraft's flagship product and core of its company-wide **GenAI platform**: designed the **system architecture** and built the **Python backend services (FastAPI)** powering Sidebar and **VerifAI** (AI contract review), driving **\$500K in new product revenue** within 3 months of private beta and **70%+ efficiency gains**.
- Engineered Sidebar's **multi-agent system** end-to-end with **LangGraph**, orchestrating specialized agents for policy Q&A, regulatory compliance, and strategic legal guidance; defined **API contracts and service integrations** with SpotDraft's Contract Lifecycle Management (CLM) platform and external legal databases.
- Built the entire **search & retrieval stack** from scratch: **semantic search**, **hybrid retrieval** (sparse BM25/BM42 + dense embeddings), relevance ranking, and **knowledge-graph-augmented RAG** over contract repositories, company policies, and regulatory databases.
- Architected a **high-throughput, fault-tolerant document-processing pipeline** at the scale of **100K to millions of documents** on **Apache Beam / Google Cloud Dataflow** for distributed processing: metadata extraction, layout-aware chunking, embedding generation, and indexing into vector and knowledge-graph stores.
- Created a **three-tier agentic evaluation framework** spanning **agent-level evals** (per-subagent correctness and tool use), **integration evals** (inter-agent handoffs), and **end-to-end evals** (final output quality), with **structured execution traces** enabling precise **failure attribution** (a stack trace for agent systems) and automated **regression detection** across releases.
- Shipped a **multi-model orchestration & routing layer** (LiteLLM) across GPT-4, Claude, and other LLMs for cost/latency/accuracy optimization with production fallbacks; **fine-tuned embedding models** for domain-specific legal retrieval and built **prompt-engineering frameworks** for reliable responses.

Senior Machine Learning Engineer

March 2025 - June 2025

Apollo.io

India

Tech: Python, OpenAI GPT, LangGraph, LangSmith, Langfuse, LiteLLM, Vector Databases, Elasticsearch, FastAPI, Docker, Ruby on Rails

- Built the **backend (Python, FastAPI)** for the **Apollo.io AI Assistant**, an agentic chatbot for lead generation, list building, and email sequencing; designed and evaluated **supervisor, mesh, and ReAct agent architectures**, setting the foundation for the company's agent platform.
- Established the **LLM evaluation practice** from scratch: built **golden datasets** and automated eval pipelines (LangSmith, Langfuse), reaching **87% coverage** and catching regressions pre-release.
- Scaled a **URL-prioritization system** for web-scale company-site scraping from 35% to **75%+ coverage** using **LLM-generated ground truth** and a custom multi-class classification model; designed and deployed a **self-hosted model-serving backend** replacing third-party services and eliminating **\$7K/month (\$84K/yr)**.

Co-Founder & CTO

January 2023 - March 2025

Stybe, AI-powered fashion e-commerce

India

Tech: Python, OpenAI GPT, Vector Databases (Qdrant), Elasticsearch, FastAPI, SQL, Docker, NLP, Computer Vision

- Owned the full AI/engineering stack as CTO: architected the **system end-to-end** and built the entire **Python backend** (FastAPI, SQL, Elasticsearch, Docker): a production **RAG system** (LLMs + vector databases) serving as an AI personal stylist for personalized fashion discovery.
- Developed **GenAI conversational search** with custom **RAG** and **hybrid retrieval** (sparse BM25/BM42 + dense embeddings), enabling natural-language product discovery with semantic understanding.

- Achieved **86%+ accuracy** in cross-platform product matching using **semantic similarity** and **vector search** (Qdrant), powering a price-comparison engine; engineered **LLM-based product tagging** (GPT-3.5) and solved the **cold-start problem** via an AI-driven implicit/explicit feedback loop for continuous personalization.

Machine Learning Engineer, Perception

April 2022 - January 2023

Motive (formerly KeepTruckin)

Remote, India

Tech: Python, C++, Go, PyTorch, Snowflake, SQL, Docker, Kubernetes

- Designed, trained, and deployed **multi-task deep-learning models** (shared-backbone architectures for object detection, multi-object tracking, and monocular distance estimation), optimized and **quantized for real-time inference** on **1M+ edge devices** (dashcams).
- Built and deployed an end-to-end **facial-recognition (Driver ID) pipeline** with backend services in **Go and Python** operating in the wild: **85%+ accuracy, 95% coverage, 100K+ daily predictions** at fleet scale.
- Trained on large-scale real-world driving datasets and applied **GAN-based synthetic data generation** (~500K frames) for domain adaptation, mitigating data bias and improving edge-case generalization.

Senior Machine Learning Engineer (Founding Engineer)

June 2021 - April 2022

XTRA (xtravision.ai)

Remote, India

Tech: Python, TensorFlow, MediaPipe, OpenAI GPT, React, Node.js, MySQL, Docker, Kubernetes, CI/CD

- Built **real-time video analytics** for online fitness classes: activity recognition via **3D pose estimation**, rep counting, and live in-browser activity metering with per-user class ranking.
- Shipped a **fitness-assessment module** (endurance and flexibility scoring with personalized recommendations) that landed **4 major clients**; leveraged **LLMs and prompt engineering** for customer-support and recommendation chatbots.

Senior Research Engineer, Machine Learning (CTO Office)

February 2020 - June 2021

Wipro Ltd.

Bengaluru, India

Tech: Python, PyTorch, Scikit-Learn, OpenCV, Flask, Java, Android, Docker, CI/CD

- Built a **CycleGAN-based document-denoising service** (perceptual loss) for noisy scans and phone captures; deployed via **Flask API** and adopted by multiple internal teams.
- Developed **one-shot object detection** for symbol spotting in architectural/engineering drawings (low-data regime), identifying **85%+ of symbols** per drawing; explored **incremental learning** (iCaRL) for continual learning.
- Created an **audio-as-vision cough-detection system** (YOLOv5 + focal loss on waveform plots, **mAP 0.84** @0.5:0.95 IoU) and a COVID-screening Android app (Mel spectrograms) with explainable predictions, **F1 0.89**.

TECHNICAL SKILLS

GenAI & LLMs: OpenAI GPT-4, Claude (Anthropic), LLaMA family, BERT, T5, Hugging Face Transformers, NLP (Natural Language Processing), Transformer Architecture Internals, Prompt Engineering, Fine-Tuning (LoRA/QLoRA), Model Distillation, Quantization

Agents, RAG & Evals: LangGraph, LangChain, LlamaIndex, LiteLLM, LangSmith, Langfuse, Multi-Agent Orchestration (Supervisor/Mesh/ReAct), Custom RAG Pipelines, Hybrid Search (Sparse + Dense), Reranking, Knowledge Graphs, Semantic Search, LLM Evaluation & Golden Datasets

Backend & System Design: FastAPI, Flask, Django, REST API Design, Distributed Systems, SQL, Caching & Search Infrastructure, System Design for Scalability & Reliability

Data, Processing & Vector Stores: Apache Beam, Google Cloud Dataflow, Snowflake, Large-Scale ETL & Document Pipelines, Qdrant, FAISS, Pinecone, Milvus, Weaviate, Elasticsearch

Languages, Cloud & MLOps: Python, JavaScript/TypeScript, Go; Azure, GCP, Docker, Kubernetes, CI/CD, Model Deployment & Serving, Inference Optimization, Edge Deployment

ML/DL & Architectures: PyTorch, TensorFlow, Scikit-Learn, OpenCV, Object Detection, Multi-Task Learning, GANs (CycleGAN), 3D Pose Estimation, Domain Adaptation

EDUCATION

College of Engineering Pune (COEP)

Aug 2017 - July 2019

M.Tech., Computer Engineering & Artificial Intelligence

Pune, Maharashtra

Dr. A.P.J. Abdul Kalam Technical University

July 2012 - July 2016

B.Tech., Computer Engineering

Ghaziabad, Uttar Pradesh

PUBLICATIONS

A Survey of Pedestrian-Autonomous Vehicle Interaction Challenges & Pedestrian Intent Identification, Springer, *Lecture Notes in Networks and Systems*, 2nd Intl. Conf. on Data & Information Sciences, 2019.

Pedestrian Safety for Autonomous Vehicles, Springer, *Advances in Intelligent Systems and Computing*, 4th Intl. Conf. on ICT for Sustainable Development, 2019.